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Genetic constraints on temporal variation of airborne reflectance spectra and their uncertainties over a temperate forest

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ABSTRACT

Remote sensing enhances large-scale biodiversity monitoring by overcoming temporal and spatial limitations of ground-based measurements and allows assessment of multiple plant traits simultaneously. The total set of traits and their variation over time is specific for each individual and can reveal information about the genetic composition of forest communities. Measuring trait variation among individuals of one species continuously across space and time is a key component in monitoring genetic diversity but difficult to achieve with ground-based methods. Remote sensing approaches using imaging spectroscopy can provide high spectral, spatial, and temporal coverage to advance the monitoring of genetic diversity, if sufficient relation between spectral and genetic information can be established.

We assessed reflectance spectra from individual *Fagus sylvatica* L. (European beech) trees acquired across eleven years from 69 flights of the Airborne Prism Experiment (APEX) above the same temperate forest in Switzerland. We derived reflectance spectra of 68 canopy trees and correlated differences in these spectra with genetic differences derived from microsatellite markers among the 68 individuals. We calculated these correlations for different points in time, wavelength regions and relative differences between wavelength regions. High correlations indicate high spectral-genetic similarities. We then tested the influence of environmental variables obtained at temporal scales from days to years on spectral-genetic similarities. We performed an uncertainty propagation of radiance measurements to provide a quality indicator for these correlations.

We observed that genetically similar individuals had more similar reflectance spectra, but this varied between wavelength regions and across environmental variables. The short-wave infrared regions of the spectrum, influenced by water absorption, seemed to provide information on the population genetic structure at high temperatures, whereas the visible part of the spectrum, and the near-infrared region affected by scattering properties of tree canopies, showed more consistent patterns with genetic structure across longer time scales. Correlations of genetic similarity with reflectance spectra similarity were easier to detect when investigating relative differences between spectral bands (maximum correlation: 0.40) than reflectance data (maximum correlation: 0.33). Incorporating uncertainties of spectral measurements yielded improvements of spectral-genetic similarities of 36% and 20% for analyses based on single spectral bands, and relative differences between spectral bands, respectively.

This study highlights the potential of dense multi-temporal airborne imaging spectroscopy data to detect the genetic structure of forest communities. We suggest that the observed temporal trajectories of reflectance spectra indicate physiological and possibly genetic constraints on plant responses to environmental change.

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1. Introduction

Remote sensing has demonstrated its potential for large-scale biodiversity monitoring (e.g. Pettorelli et al., 2014; Turner, 2014; Skidmore et al., 2015; O'Connor et al., 2015; Navarro et al., 2017; Skidmore et al., 2021). It is especially valuable in the context of current environmental change affecting global biodiversity decline (Proenca and Pereira, 2013; Blowes et al., 2019), as it overcomes spatial and temporal limitations of ground-based assessments of biological variation (Jetz et al., 2016; Gamon et al., 2019). Remote assessments of functional diversity (Asner and Martin, 2009; Singh et al., 2015; Wang et al., 2019; Schimel et al., 2019; Schneider et al., 2017; Zheng et al., 2021) may advance our understanding of ecosystem functioning and may be used for ecological modeling. Genetic diversity is an equally important facet of biodiversity, but has received less attention in remote sensing studies.

Genetic diversity is a key component of biodiversity loss (Wilson and Peter, 1988) and has been defined as an Essential Biodiversity Variable (EBV) (Pereira et al., 2013; Skidmore et al., 2021). Greater genetic variation reflects the availability of more alleles, or versions of genes, resulting in a wider range of phenotypic responses and thus increased potential to maintain ecosystem functioning and evolutionary processes under changing environmental conditions (Hughes et al., 2008; Bolnick et al., 2011; Des Roches et al., 2018; Moritz, 2002). Monitoring current genetic diversity and its distribution across populations of interbreeding individuals, and understanding what causes this to change, is essential for preventing further biodiversity loss. Measurement of allele pools is commonly based on field sampling followed by laboratory analyses, which both require substantial time and effort per sample. Thus, the assessment of genetic diversity would benefit from approaches that overcome scale-induced limitations.

Remote sensing is being used to assess phylogenetic diversity in plant communities in direct (Cavender-Bares et al., 2016; Cavender-Bares, 2019; Lin et al., 2021) and indirect ways (Schweiger et al., 2018; Frye et al., 2021; Rocchini et al., 2010; Feret and Asner, 2012). These assessments are done at species or higher taxonomic levels, while assessments of intraspecific genetic variation are more challenging because trait differences expressed in spectra are likely to be less pronounced at the within- than at the between-species level (Hulshof and Swenson, 2010; Guillén-Escribà et al., 2021). Dense temporal coverage of imaging spectroscopy acquisitions may allow detection of more differences among individuals — thus helping to resolve intraspecific variation — if traits expressed at different times or ontogenetic stages are underpinned by different genes within an individual and if thus a link between temporally variable spectral and temporally constant genetic variation can be established (Yamasaki et al., 2017; Czyż et al., 2020).

Remote sensing allows for systematic monitoring of the land surface on a wide range of temporal scales. Time series of remotely sensed data reveal information about the dynamic behavior of vegetation and have significant advantages over single time-point measurements (Goetz, 2009), which tend to be a limitation of ground-based investigations (Albert et al., 2011). Repeat-visit airborne imaging spectroscopy data can, for example, capture seasonal changes in vegetation spectra. Changes in spectra over time may reflect leaf green-up (Garcia and Ustin, 2001), leaf ontogenetic stages (Chavana-Bryant et al., 2017), mesophyll development (Rivera et al., 2002), seasonal changes in equivalent water thickness (Roberts et al., 1997), lignin and cellulose turnover (Patton and Gieseker, 1942), and start of senescence (Myneni et al., 1997). Reflectance spectra can be used to assess plant water stress (Suárez et al., 2008), and pigment (Ustin et al., 2009) and nitrogen concentration in vegetation (Ollinger et al., 2002; Oldeland et al., 2010). These are all traits, or observable features of individuals, which may be genetically constrained, meaning that they have a genetic basis and thus their variation may in part be explained by genetic variation. Therefore, time-resolved imaging spectroscopy data may help to remotely sense genetic variation even within a species, as it has already been shown to support the identification of vegetation types (Merton, 1998) and plant species (e.g. Castro-Esau et al., 2006; Asner et al., 2006; Somers and Asner, 2013). Dense temporal coverage allows the assessment of plant responses to environmental conditions, which drive adaptation in species (Hut et al., 2012). Some adaptations may be observed in so-called temporal niche partitioning by individuals with varying strategies defined by traits, and relationships between traits, which vary in different ways over time (Angert et al., 2009; Levine and HilleRisLambers, 2009).

Dense multi-temporal imaging spectroscopy data with their rich spectral information, can simultaneously and systematically assess multiple plant traits and relations among them. Relationships between plant traits are recognized across levels of biological organization (Joswig et al., 2021). On the elementary level, stoichiometry explains the balance among chemical elements as the basic components of living organisms (Sterner and Elser, 2002). The specific composition of chemical elements in plants, in the form of molecular building blocks but also as nutrients and cofactors, is related to their morphological and physiological traits. These are in turn related to each other especially in terms of traits describing size or allometry, versus economic investments of nutrients over a plant's lifetime (eg. Wright et al., 2004), and describe a spectrum of form and function across plants (Díaz et al., 2016). For example, carbon fixed by photosynthesis is allocated differently into shoots and roots, with trade-offs between these (Tilman, 1988), and may be used for energy or energy storage in the form of nonstructural carbohydrates, for structural changes like lignification, or defensive measures such as phenol production (Eichenberg et al., 2015). The set of traits ultimately expressed by individuals builds on specific ecological strategies of plants (Westoby, 1998) and affects ecosystem processes (Field et al., 1986; Marks and Lechowicz, 2006; Wright et al., 2003; Reich et al., 1998; Niinemets, 2001). Natural selection favors strategies which optimize fitness in a given environment, some of which vary within species (He et al., 2009; Albert et al., 2011), while others are characteristic of phylogenetic groups (eg. Westoby and Wright, 2006; Kerkhoff et al., 2006), or of convergent evolution across phylogenies (Gamon et al., 2019; Ustin et al., 2009; Reich et al., 2003). Expressed strategies characteristic for plant phylogenetic groups could be derived by combining information derived from spectral indices such as e.g. NDVI (*Normalized Difference Vegetation Index*, Rouse et al., 1974), NDWI (Normalized Difference Water Index, Gao, 1996) or PRI (Photochemical Reflectance Index, Gamon et al., 1992), partial least squares regression (PLSR) models (e.g. Singh et al., 2015) or potentially also the relationships within the full range of reflectance spectra.

In this study, we present a proof of principle for the use of dense multi-temporal airborne imaging spectroscopy data to investigate intraspecific genetic variation of temperate forest trees. We link multitemporal aerial imaging spectroscopy with data on the genetic structure of a *Fagus sylvatica* L. (European beech) population derived from five nuclear microsatellites. We hypothesize that genetically similar individuals express similar phenotypes as detected in reflectance spectra at different time points. We investigate (1) which of the time-resolved spectral feature(s) reveal the highest similarity with genetic structure, (2) to what extent relationships between spectral features are genetically constrained, and (3) under which environmental conditions and at what temporal scales genetic-spectral similarities are largest. Further, we suggest underlying physiological processes that may correlate with genetic structure, and we investigate to what extent the uncertainties associated with the spectral signal propagate to correlations.

We take advantage of detailed in situ and airborne mapping of trees at the study site (Morsdorf et al., 2020). We use genetic markers of intraspecific genetic variation consisting of lengths of five nuclear microsatellite markers, long-term measurements of environmental variables at the study site, and available high-resolution spectral data required for time-series analysis of vegetation spectral measurements. Furthermore, we account for inherent uncertainties of spectral measurements (Schaepman et al., 2015; Schaepman and Dangel, 2000) and propagate them through the analysis. This approach enables us to assess

reliability of information derived from spectra (Woolliams et al., 2014), which is of importance in the context of multi-temporal investigations and coordination of upcoming imaging spectroscopy missions (Thompson et al., 2020) and retrievals (Cawse-Nicholson et al., 2021). With that, we hope to contribute to developing scalable approaches to monitor and understand biodiversity change on its basic, intraspecific genetic level.

2. Materials and methods

2.1. Study area

The study area ('Lägern') covers 12.6 ha of semi-natural temperate mixed forest located on an up to 60°-steep south-facing slope of the Lägern mountain in Switzerland (47°28′N, 8°21′E). Its elevation ranges from 620 to 810 m a.s.l. The climate of this region is characterized by a mean annual temperature of 7.4 °C and mean annual precipitation of 1,000 mm (Etzold et al., 2011). The vegetation cover is classified as Temperate Deciduous Broadleaf Forest (UNEP-WCMC, 2000) with 13 tree species including three conifers and ten angiosperms, with *Fagus sylvatica* L. (European beech) being the most abundant (Guillén-Escribà et al., 2021). Mean height of canopy trees at the site is 30.6 m and the tree age ranges between 53 and 185 years, with a maximum diameter at breast height of 150 cm (Eugster, 2021). The canopy is mostly closed and characterized by a complex vertical structure (Schneider et al., 2017). The study area is located in an unmanaged part of the forest and has been a forest ecosystem research site for the last four decades (Kloeti et al., 1989). Individual trees have been reconstructed in 3D using ground and airborne laser scanning, and simulated using 3D radiative transfer models, allowing the validation of airborne data with high accuracy (Schneider et al., 2014; Schneider et al., 2017; Kükenbrink et al., 2021). The tree trunks were previously georeferenced using a differential GPS system and tied to reference points of the Swiss national grid (LV95), with a polygon traverse and additional fixed points to measure trunk locations in the challenging terrain (Leiterer et al., 2015). Because the slope geometry distorted the nadir projection of crowns to trunks, tree crowns were mapped separately and linked to tree trunks using photogrammetry (Guillén-Escribà et al., 2021; Torabzadeh et al., 2019). The mapped tree crowns of the site allowed us to assign reflectance spectra of multiple pixels and genetic information to 68 analyzed *F. sylvatica* individuals.

2.2. Spectral data

The spectral dataset contains 87 acquisitions of the Airborne Prism Experiment (APEX) Airborne Imaging Spectrometer (AIS) (Schaepman et al., 2015) obtained between 2009 and 2019 and covering the range of 23–1160 Cumulative Growing Degree Days [°C] (CGDD) (Table S1). Preprocessing of the raw APEX data including radiometric calibration and correction for spectral shifts was done using the APEX processing and archiving facility (Schaepman et al., 2015; Hueni et al., 2008; Hueni et al., 2012; Richter et al., 2010). The adjustment for smile effects and atmospheric compensation for surface reflectance accounting for solar zenith and azimuth and viewing angles was done using ATCOR, and the reflectance data were georectified in PARGE (Schläpfer and Richter, 2002; Hueni et al., 2017). This resulted, for each acquisition, in a georectified reflectance spectra dataset containing 284 spectral bands with a variable spectral resolution of 7–13.5 nm in the wavelength range from 399 to 2425 nm, with 2 m spatial resolution at canopy level. For the analyses, we selected only acquisitions with NDVI >0.5 and with the area of interest being fully covered. This resulted in a final set of 69 cloud-free acquisitions obtained during 27 airborne missions acquired under leaf-on conditions (Fig. 1).

Along with the reflectance spectra data, we obtained data about radiometric uncertainties. The uncertainties were calculated for each pixel and each spectral band based on the radiometric calibration of APEX (Schläpfcr et al., 2000; Schläpfer and Schaepman, 2002; Jehle et al., 2014; Hueni et al., 2017). They account for system noise derived from multiple measurements using an integrating sphere, pressure changes affecting the spectral positions of derived bands, sensitivity to sensor temperature, smear effect, spectral band binning, electronic anomalies, and estimation errors of radiometric gain and offset (Hueni et al., 2017).

For analysis, we derived reflectance spectra and radiometric uncertainties of pixels that represent the 68 canopy individuals of *F. sylvatica*. For each tree, we selected only those pixels whose centroid lay within the area of the mapped tree crown (see section 2.1) and we used average values of reflectance spectra weighted by their inverse uncertainty per tree individual per day (see section 2.6).

2.3. Genetic data

Relatedness of individuals was assessed using five nuclear microsatellite markers, which are repetitive DNA sequences commonly used for population genetics, from the 68 canopy individuals of *F. sylvatica* as detailed in a previous study (Czyż et al., 2020). Briefly, we sampled leaf disks (diameter 1.15 cm) from each tree in September 2013 and stored them on silica gel. Genomic DNA from the sampled material was extracted using the cetyl trimethylammonium bromide (CTAB) method following the procedure of Doyle (1990). From the extracted DNA, we amplified five highly variable microsatellite loci (FS1–03, FS1–15, FS3–04, FS4–46, FCM5; Pastorelli et al., 2003). To assess the degree of polymorphism at each microsatellite locus, we performed capillary electrophoresis on an ABI-3720 sequencer (Thermo Fisher Scientific, UK) and we used GeneMapper software (Applied Biosystems) to determine the length of the analyzed microsatellites for each sampled tree

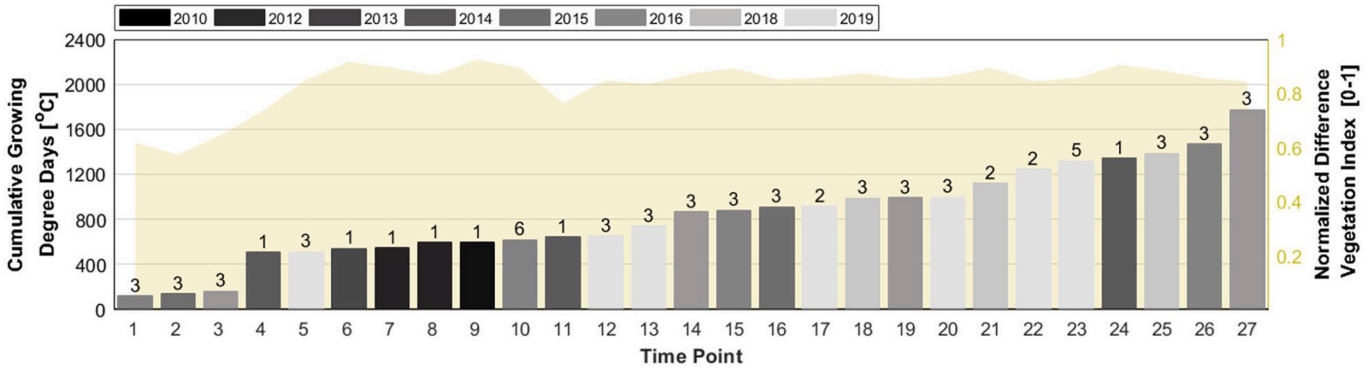


Fig. 1. Airborne imaging spectroscopy missions segregated into time points based on Cumulative Growing Degree Days [°C] (CGDD) and shaded by year of acquisition. Each CGDD bar corresponds to a single day of acquisition. A base temperature of 5 °C was used (see section 2.4). The numbers on top of the bars specify the number of acquisitions at that day (multiple acquisitions per day). The background represents the NDVI calculated for each time point.

(Table S2).

2.4. Climatic data

We used climatic data collected on a flux tower 47 m vertically above ground level at the site (Eugster, 2021). The data cover the 11-year period 2009–2019 representing measurements at 30-min resolution.

The data used in the present study included air temperature [°C] and Vapor Pressure Deficit (VPD) at different temporal scales as follows. For each airborne acquisition, we calculated the mean temperature (TMP) and mean Vapor Pressure Deficit (VPD) of the day, aggregated temperature (11TMP) and aggregated Vapor Pressure Deficit (11VPD) over 11 consecutive days before the acquisition, and Cumulative Growing Degree Days (CGDD) and Cumulative Growing Vapor Pressure Deficit Days (CGVPD). The moderate time scale of 11 days was selected based on a relatively high impact of selected climatic variables on spectral-genetic similarities across all the spectral bands at this time span (Supplementary, *'Selection of moderate time scale'*). The CGDD were calculated with a base temperature of 5 °C (Polgar and Primack, 2011). Additionally, we included the Day of the Last Spring Frost (DLSF) [DOY] and Last Year Climatic Water Balance of March–August (LYCWB) calculated as monthly potential evapotranspiration (Hargreaves, 1994) subtracted from the monthly sum of precipitation.

2.5. Spectral-genetic similarity estimation

To investigate the correlation between reflectance spectra and genetic structure, we conducted partial Mantel tests between spectral and genetic distances among the 68 individual trees. We refer to the derived correlations as spectral-genetic similarities.

Firstly, we calculated the pairwise distances between trees using (1) the spectral Euclidean distance of each spectral band from $i=1$ to $i=284$ for each of the 27 time points (average of multiple pixels and multiple acquisitions per tree per day, see 2.2 and 2.6), (2) the spectral Euclidean distance of relative differences, d_n between wavelengths calculated for all pairwise combinations of spectral bands for each time point as (eq. 1):

$$d_n = \frac{|x_i - x_j|}{x_i + x_j} \tag{1}$$

where x is the reflectance value of the wavelength at spectral bands i and j respectively, (3) the spatial Euclidean distance based on the geographical position of the stem bases of the individuals, and (4) Nei's genetic distance representing the relationship between individuals based on their genetic divergence from a common ancestor, calculated based on the length of analyzed microsatellites of each *F. sylvatica* individual. We decided to introduce relative differences between wavelengths, because this normalization (1) reduced the impact of brightness, especially important in multi-temporal analysis, which entails variation in sun illumination and geometry of acquisition, and (2) simultaneously accounted for interactions among multiple traits that affect the spectral signatures of analyzed trees and may be indicative about their biological strategies. Nei's genetic distance is used to compare populations where drift and mutation may explain genetic differences (Nei, 1972) and we calculated it using the *nei.distance* function of the R package *poppr* (Kamvar et al., 2014, R v.8.5). This resulted in (1) 284 × 27, (2) (284×(284–1)/2) × 27, (3) one and (4) one symmetric matrices of dimension 68 × 68 representing pairwise distances between the 68 trees.

We then performed partial Mantel tests (Mantel, 1967; Smouse et al., 1986) to derive the spectral-genetic similarities between each spectral distance matrix of each time point and the genetic distance matrix. We assume that trees which are spatially closer together may be exposed to a more similar environment and we thus controlled for spatial distances between trees at the study site by subtracting the correlation of residuals from the linear regression of spectral-spatial correlation and genetic-spatial correlations (Smouse et al., 1986). We calculated partial Mantel correlations in Matlab v. R2017b software.

Finally, we calculated to what extent variation in spectral-genetic similarities was explained by variation of different environmental variables at different temporal scales. For that, we calculated the Pearson correlations of time point-specific spectral-genetic similarities and time point-specific environmental variables that represented daily, multi-daily, seasonal and inter-seasonal environmental conditions (Fig. 2).

For clarity of representation, we do not visualize the spectral-genetic similarities lower than 0, as such representa negative dependency of spectral signatures on genetic structure (genetically more related individuals being more dissimilar spectrally at these regions), which can not be used to infer genetic constraints. In our analysis, negative spectral-genetic similarities account for 30.7% of results mean of negative correlations −0.05 and lowest value −0.15 with $p = 1$. Furthermore, we discuss only the spectral-genetic similarities and their correlations with environmental variables, which are significant at $p < 0.05$ and of which the relative uncertainties do not exceed 20%.

2.6. Uncertainty propagation

Each analyzed tree individual in the spectral data comprises multiple pixels represented as surface reflectance values with their uncertainties of radiance measurements (see 2.2). We accounted for the uncertainties as relative reliability of measurements and propagated them to the spectral-genetic similarities and their correlations with environmental variables. More specifically, we incorporated the uncertainties throughout the analysis at two stages. In the first stage, we applied pixel-based uncertainty, u_i (where i stands for pixel within a tree crown) to derive the mean weighted by the inverse uncertainty, $\bar{x}_t$ (eq. 2) of spectral reflectance with its uncertainty, $u_{\bar{x}_t}$ (eq. 3) of each tree crown per acquisition.

$$\bar{x}_t = \sum \left(\frac{x_i}{u_i^2} \Big/ \sum_i \frac{1}{u_i^2} \right) \tag{2}$$

$$u_{\bar{x}_t} = \sqrt{\frac{1}{\sum_i u_i^2}} \tag{3}$$

Following the same operations, we calculated the mean weighted by the inverse uncertainty, $\bar{x}_d$ (eq. 4) of spectral reflectance with its uncertainty, $u_{\bar{x}_d}$ (eq. 5) of each tree crown per day as the unit of further analyses:

$$\bar{x}_d = \sum \left(\frac{\bar{x}_{t_j}}{u_{\bar{x}_{t_j}}^2} \Big/ \sum_t \frac{1}{u_{\bar{x}_{t_j}}^2} \right) \tag{4}$$

$$u_{\bar{x}_d} = \sqrt{\frac{1}{\sum_j u_{\bar{x}_{t_j}}^2}} \tag{5}$$

where j stands for the acquisition within a day sorted by CGDD.

In the second stage, we propagated the uncertainty of each tree crown per day to derive uncertainties of spectral-genetic similarities and their correlations with environmental variables. For that, we followed the law of propagation of uncertainties (CIPM; JCGM, 2008), assuming the uncertainties to not co-vary, to sequentially calculate absolute uncertainties of pairwise Euclidean distances, z-score of Euclidean distances, correlation coefficients, partial Mantel correlations, relative differences, means of partial Mantel correlations, standard deviations of partial Mantel correlations. The detailed description of the uncertainty propagation as sensitivity analysis of each function to the uncertainty of its driver is described in the supplementary data (Supplementary, *'Propagation of uncertainties'*). We represent the derived uncertainties as relative uncertainties to the spectral-genetic similarities and their

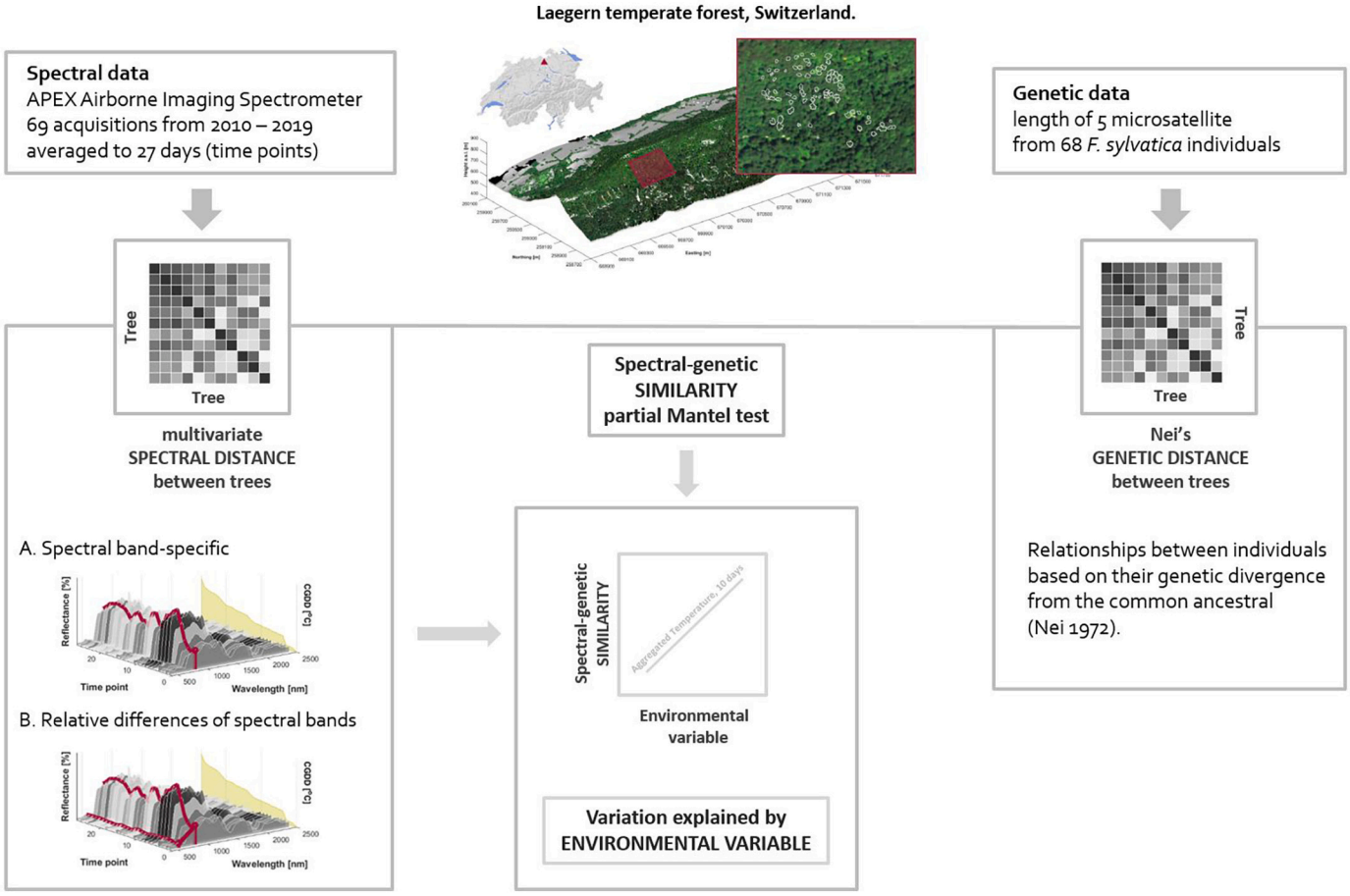


Fig. 2. Schematic representation of the workflow. Spectral-genetic similarities were calculated as similarities between distance matrices derived from multi-temporal spectral and genetic data. The spectral data were considered as a single spectral band and as relative differences between the spectral bands, represented with one and two red lines overlaid on spectral responses averaged per day and shaded by year of acquisition. The Cumulative Growing Degree Days [°C] (CGDD) of each acquisition is represented with green shaded area. The spectral-genetic similarities were finally correlated with environmental variables. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

correlations with environmental variables.

3. Results

3.1. Spectral-genetic similarities

The similarities between reflectance spectra and genetic structure (partial Mantel statistic r) varied across single spectral bands and time points (Fig. 3). The highest spectral-genetic similarity reached 0.33 (relative uncertainty: 5%, $p = 0.01$) for a spectral band centered at 1925 nm at the 19th time point (993 CGDD, July) and was 36% higher than the highest result (0.21) derived from analyses where the mean reflectance spectra per tree were not weighted by the measurement uncertainty (not shown). Across all time points, the spectral-genetic similarities derived from the spectral regions ranging from 650–680 nm and 1950–2000 nm were relatively stable over time, the rest differed among time points. Across the whole spectrum, the 14th (866 CGDD, July) time point had the highest (0.10) spectral-genetic similarities. The beginning of the growing season (116–515 CGDD, April untill beginning of July) was characterized by relatively low spectral-genetic similarities for spectral bands ranging from 1100–2500 nm. Towards the end of the growing season (1317–1776 CGDD, end of July untill September) spectral-genetic similarities were relatively high for spectral bands ranging from 680–1800 nm. Time points which included ~~of~~ at least five acquisitions — the 10th and the 23rd (see Fig. 1) — had higher spectral-genetic similarities derived from spectral bands ranging from 550–650 and 690–1350 nm, respectively. Additional high spectral-genetic similarities occurred at particular time points: at the second time point~~,~~ for spectral bands ranging from 400–500 nm; at time point 14, for spectral bands around 500 nm; at the time points 10, 12, 20 and 27 for spectral bands around 500–600 nm; at time point 26 for spectral bands around 600–680 nm, and time point 14 for spectral bands around 680–1350 nm, at time point six for spectral bands around 1400–2300 nm and at time points 26 and 27 for spectral bands around 2350–2450 nm. Across the spectrum, the lowest spectral-genetic similarities overtime were found for spectral bands ranging from 400–420, 670–680, 1350–1400 and 1900–2000 nm, which are the spectral bands affected by the scattering and absorption features in the atmosphere and also spectral regions having the largest relative uncertainty. The overall relative uncertainty of spectral-genetic similarities derived from single wavelengths had similar patterns over time and increased at longer wavelengths.

In comparison to similarities between reflectance spectra and genetic structure derived from single spectral bands, spectral-genetic similarities derived from relative differences between spectral bands were on average lower (values of partial Mantel r of −0.002 for relative differences between spectral bands vs 0.03 for single spectral bands), but reached larger value and were moderately more stable (mean standard deviation across time points of 0.06 for relative differences between spectral bands vs 0.07 for single spectral bands) (S. Fig. 2). We found the highest spectral-genetic similarity (0.40) for relative differences for a spectral band centered at 515 nm with a spectral band centered at 1925 nm (relative uncertainty: 2.9%, $p = 0.001$) during the 19th time point

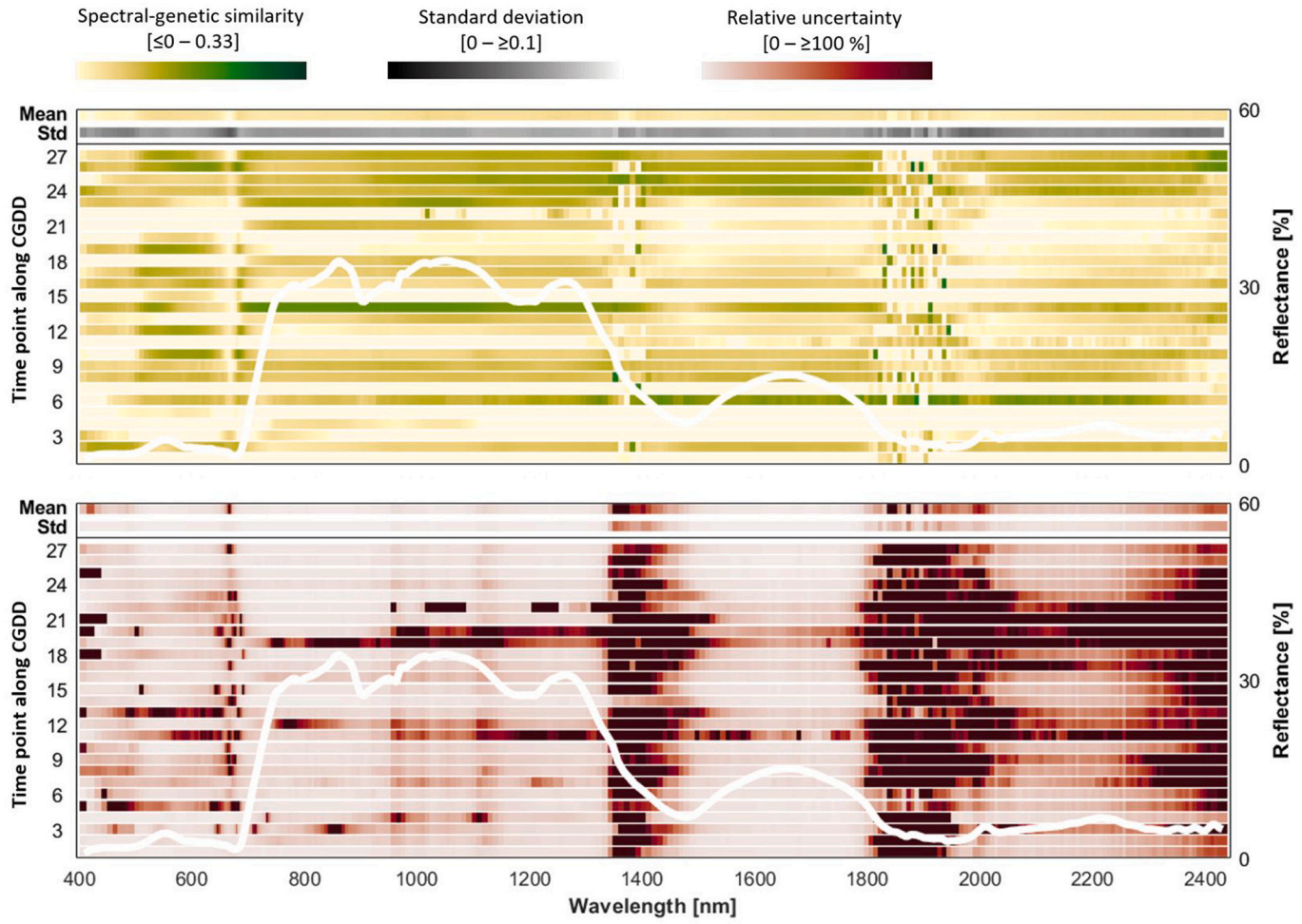


Fig. 3. Spectral-genetic similarities (values of partial Mantel r) for the population of *F. sylvatica* derived from spectral bands for each time point segregated according to CGDD. Additionally, the mean and standard deviation of the similarities from all time points are shown at the top of the panel. The shown values range from ≤0 (white) to 0.33 (dark green). The relative uncertainties of the spectral-genetic similarities for all the time points across spectral bands are represented on the lower panel. The white lines represent the mean canopy reflectance of all trees. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

(993 CGDD, July) (S. Fig. 3). The relative uncertainties of spectral-genetic similarities derived from relative differences are on average higher (8%) than relative uncertainties of spectral-genetic similarities derived from single wavelengths (1.8%).

3.2. Spectral-genetic similarities explained by environmental variables

Correlations between spectral-genetic similarities and environmental variables showed different patterns on different temporal scales (Fig. 4). Daily mean temperature (TMP) was positively correlated with spectral-genetic similarity derived from spectral bands above 1000 nm and around 1900 nm, which is highly affected by water absorption. The spectral-genetic similarities derived from around the 1900 nm spectral range were also positively correlated with the vapor pressure deficit of the day of the acquisition (VPD). When considering environmental conditions over a longer time period, the correlations between environmental variables and spectral-genetic similarities were not significant for Aggregated Temperature over 11 consecutive days before the acquisitions (11TMP) and were partly significant for Aggregated Vapor Pressure Deficit over 11 consecutive days before the acquisitions (11VPD). 11VPD was negatively correlated with spectral-genetic similarities derived from spectral bands around 650–680 nm and 1900–2100 nm. On the seasonal scale, the Cumulative Growing Degree Days (CGDD) and Cumulative Growing Vapor Pressure Deficit Days (CGVPDD) showed similar patterns of positive correlation with spectral-genetic similarities above 450 nm; however, the CGDD showed a stronger correlation on this time scale than the CGVPDD, which in contrast has higher impact on spectral-genetic similarities derived from the visible part of the spectrum (400–700 nm). The Day of the Last Spring Frost (DLSF) was non-significantly ($p > 0.05$) negatively correlated with the spectral-genetic similarity derived from all the spectral bands above 680 nm. Similarly, at the longest investigated temporal scale represented by the Last Year Climatic Water Balance (LYCWB), spectral-genetic similarities were non-significantly ($p > 0.05$) correlated with this environmental variable for all spectral bands.

In comparison to the correlation between climatic variables and spectral-genetic similarities derived from single spectral bands, correlations between climatic variables and spectral-genetic similarities derived from relative differences between spectral bands were more pronounce at the shortest (TMP and VPD) and longest time scales (DLSF and LYCWB) (S. Fig. 4). For both TMP and VPD, we found significant negative correlations, with spectral-genetic similarities derived from relative differences of spectral bands ranging from 400–420 nm, with spectral bands ranging from 1500–1800 nm. At the longest considered time scale, the spring frost (LDSF) and water balance of the preceding year (LYCWB) were, respectively, positively and negatively correlated with spectral-genetic similarities derived from relative differences of spectral bands ranging from 1000–1350 nm, with spectral bands ranging

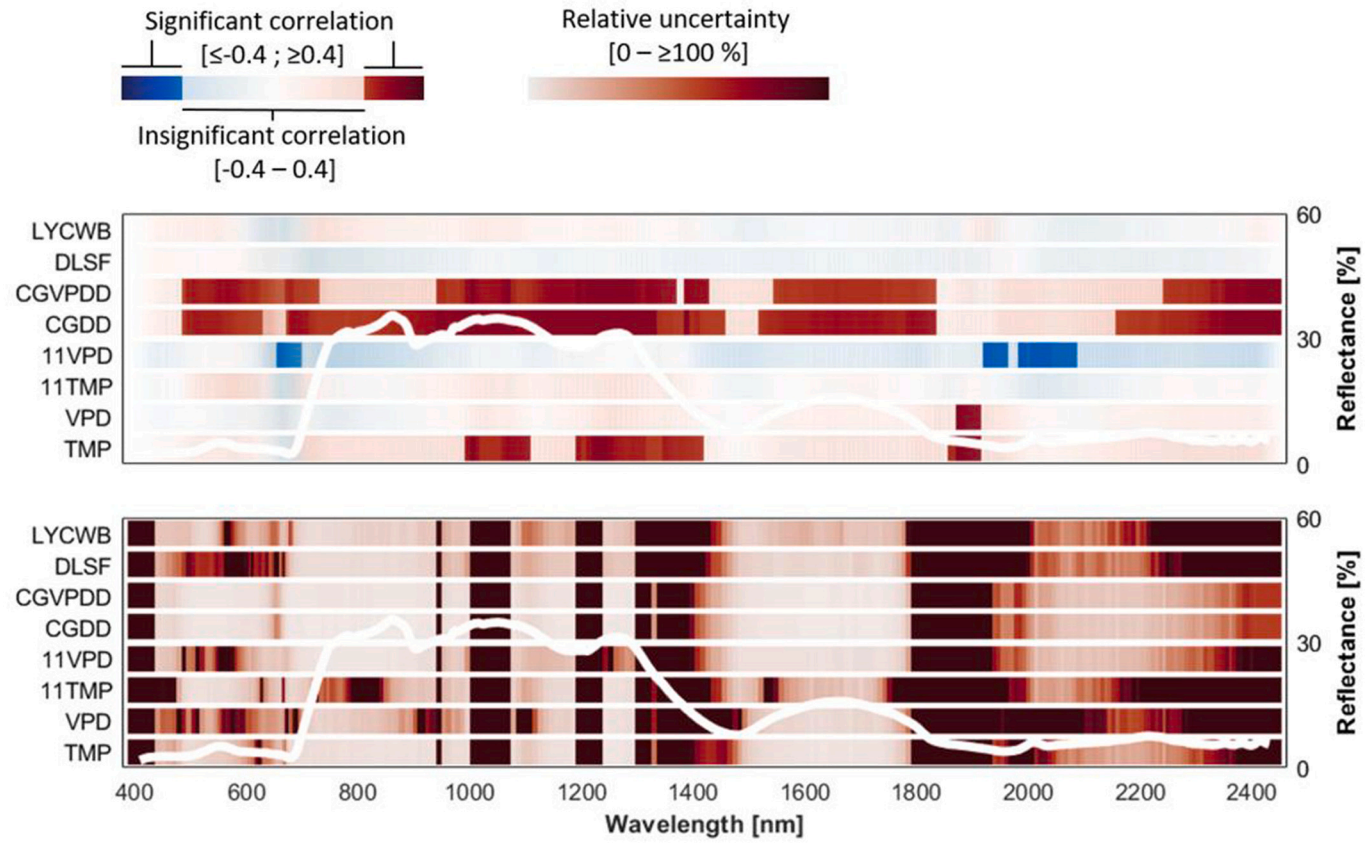


Fig. 4. Correlations between spectral-genetic similarities derived for single spectral bands and: temperature of the day of the acquisition [°C] (TMP), Vapor Pressure Deficit of the day of the acquisition [%] (VPD), Aggregated Temperature over 11 consecutive days [°C] (11TMP), Aggregated Vapor Pressure Deficit over 11 consecutive days [%] (11VPD), Cumulative Growing Degree Days [°C] (CGDD), Cumulative Growing Vapor Pressure Deficit Days [%] (CGVPDD), Day of the Last Spring Frost (DLSF), Last Year Climatic Water Balance (LYCWB). The shown values range from ≤ 0.6 (dark blue) to the ≥0.6 (dark red). The bottom panel shows the relative uncertainties of correlations between spectral-genetic similarities and environmental variables. The white lines represent the mean canopy reflectance of analyzed trees. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

from 1500–1800 nm.

The relative uncertainties of the correlation of environmental variables with spectral-genetic similarities were on average higher for results derived from relative differences of spectral bands than for results derived from single spectral bands, and were higher at longer wavelengths, for spectral bands for which the correlations were weaker, and for spectral bands around 1000 and 1200 nm.

4. Discussion

4.1. Spectral-genetic similarities suggest underlying biochemical and morphological traits

The observed similarities between information derived from airborne imaging spectrometer data and the genetic structure of a

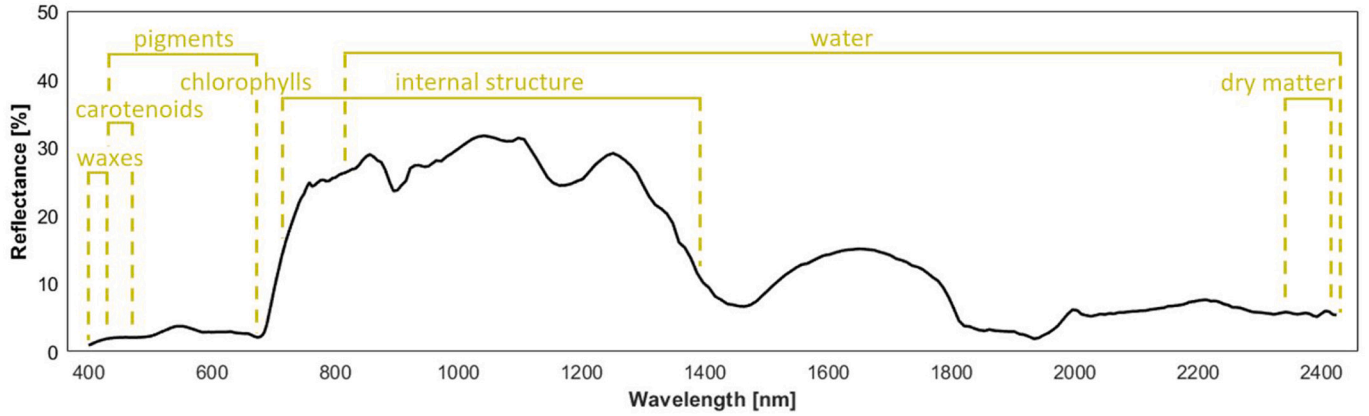


Fig. 5. The revealed spectral features and some underlying biochemical and morphological traits indicative about the genetic structure of the F. sylvatica population over time of spectral acquisitions. (The spectral ranges are limited to results of our analysis and do not represent full spectral coverage influenced by particular vegetation traits after: Long et al., 2003; Gitelson et al., 2006; Moss and Loomis, 1952; Peñuelas et al., 1994; Gausman et al., 1971; Carlson et al., 1971; Peterson et al., 1988).

population of *F. sylvatica* vary among the time points and across the spectrum for both single spectral bands, and relative differences between spectral bands. However, some reflectance spectra are similar for genetically similar individuals more systematically across all time points (Fig. 5). The spectral regions that express on average high, but not necessarily stable, spectral-genetic similarities lay in the range of 450–680 nm, corresponding to leaf pigment content and composition (eg. Moss and Loomis, 1952), 1000–2300 nm, corresponding to leaf water content that influences the reflectance spectra above 800 nm (Carlson et al., 1971), and 2350–2450 nm, affected by dry matter content (Peterson et al., 1988). Additional spectral-genetic similarities that are indicative of genetic structure and relatively time stable were derived in the 700–1400 nm spectral range. This spectral region is dominated by internal leaf scattering of light, related to the mesophyll structure and thus intercellular air spaces (Gausman et al., 1971), that are accentuated in canopy-level measurements by scattering within the tree canopy (Ollinger, 2011). Beyond similarities derived from these spectral regions, there are similarities between spectral information and genetic structure specific to different time points. At some time points, the spectral region ranging from 420–450 nm related to the leaf carotenoids content (eg. Gitelson et al., 2006) showed similar reflectance values for genetically similar individuals. The absorption feature centered at 680 nm, related to chlorophyll content (eg. Peñuelas et al., 1994), represents low spectral-genetic similarities, which were also stable over time for single spectral band investigations. The relative differences between spectral bands related to carotenoids and leaf waxy layer, influencing the reflectance spectra ranging from 400–420 nm (eg. Long et al., 2003) with spectral bands related to the leaf water content (1400–1900 nm), are on average highly indicative of the genetic structure of the studied population. Strong spectral-genetic similarities at particular time points may occur if a genetically determined trait is only expressed under particular environmental conditions occurring at that time point or if a trait is only expressed at a particular developmental stage of the tree. For example, features of reflectance spectra during pigment breakdown in leaves (Matile et al., 1999), affected by genetic variation between individual trees, can only be assessed before leaf shedding.

4.2. Genetically constrained traits and the temporal environments in which they are expressed

Because many genetically determined traits are only expressed under particular environmental conditions (Falconer et al., 1996), the inclusion of environmental variables over different time scales may help to explain observed links between genetic structure and remotely sensed phenotypes of the *F. sylvatica* population. On a day-to-day time scale, higher daily temperature leads to stronger correlation of reflectance measured at single spectral bands with genetic structure for spectral regions linked with leaf water content (above 800 nm). That may suggest that the water balance in the population is related to genetic structure when the current air temperature is higher. In comparison, the relationships between spectral bands influenced by the waxy layer (400–420 nm) and water content (1500–1800 nm) follow the genetic structure of the population when the vapor pressure deficit is lower and the ambient air is colder. This may indicate that the correlation between genetic structure and photoprotection via the waxy layer (Long et al., 2003) is more apparent (Shepherd and Wynne Griffiths, 2006) when plants are exposed to humid and cold air, differently tolerated by genetically more distant individuals. The leaf waxes provide an important barrier to water loss, and reduce light penetration into the leaf by specular reflection (Vanderbilt and Grant, 1985; Petibon et al., 2021), which may relate to mechanisms related to the genetic structure. Finally, analyses on the day-to-day time scale suggest that genetically constrained traits are more correlated with daily temperature than with daily vapor pressure deficit. However, more genetically constrained traits seemed to be strongly expressed under different 11-day vapor pressure deficit conditions (11VPD) than different 11-day temperature conditions (11TMP). A lower vapor pressure deficit for 11 days before the acquisition led to observation of higher genetic-spectral similarities of the analyzed population over the whole spectrum, especially the chlorophyll content (680 nm) and water content (1900–2100 nm). This suggests that the balance between water management and light absorption by pigments is more similar for genetically similar individuals under favorable environmental conditions sustained at moderate time scale (11 days). Other studies have shown that water relations of *F. sylvatica* can vary within the species (Gonzalez de Andres et al., 2021) and could be characteristic for genetically different provenances (Peuke et al., 2002). Our observations indicate that water balance may correspond more to genetic structure when the water availability at 11-day time scale is not a limitation — at least within this single population over the observed time points. However, we treat the correlations based on results derived from water and carbon dioxide absorption features (1900–2050 nm) with caution, as the uncertainties presented in our analysis do not account for uncertainties due to atmospheric correction. More specifically, that absorption due to these substances along the path from leaves and canopies, through the atmosphere, to the aerial sensor might not be perfectly accounted in atmospheric compensations.

Across the season, our analyses found that the influences of water availability as measured by CGVPDD and of temperature as measured by CGDD followed a similar pattern and held positive correlations with spectral-genetic similarities derived from single spectral bands for spectral regions related with pigment composition and content (400–680 nm) and water content (above 800 nm). Compared with CGVPDD, CGDD correlated also significantly with spectral-genetic similarities derived from 700–1000 nm, related to canopy scattering properties. This suggests that chlorophyll content (absorption feature at around 680 nm), leaf internal structure (700–1000 nm) and water balance (800–2390 nm) correlate with genetic structure towards the end of a hot and dry growing season. We furthermore speculate that this may be due to more similar foliage development, thus capacity to harvest light, of genetically similar individuals. Finally, the leaf internal structure as inferred from the spectral bands ranging from 1000–1400 nm correlated with water content as inferred from the spectral bands ranging from 1500–1800 nm may be informative about the genetic structure of the population when the last day of the spring frost was relatively late and the year proceeding the acquisition was relatively dry. This was indicated by the positive correlation with day of last spring frost (DLSF) and negative correlation with Last Year Water Climatic Balance (LYWCB), and the spectral-genetic similarity of this spectral range (S. Fig. 4). We suggest that there could be a similarity in closely related individuals in their responses to frost and drought, which in turn affects leaf morphology and development (Menzel et al., 2015). It has been demonstrated that juvenile *F. sylvatica* individuals from genetically different provenances vary in sensitivity to frost damage (Kreyling et al., 2012) and the development of leaf buds may be affected by water availability.

Together, our observations show that the spectral information derived from airborne imaging spectrometer data can be indicative for the genetic structure of the population of a dominant tree species and that more genetic structure can be revealed if spectral information is obtained under multiple environmental conditions and over time, which allows plants to express more of their genetically constrained traits than under a single environmental condition or at a single time of development.

4.3. Limitations and outlook

In our study, we focused on observing the temporal dynamics of reflectance spectra, on how those spectra correlate with genetic structure of a tree population, and under which environmental conditions correlations for particular spectral features with genetic structure are particularly strong. The environmental variables we correlated were

water availability as measured by either vapor pressure deficit or climatic water balance and temperature on multiple temporal scales. Both of these variables strongly affecting *F. sylvatica*. The investigation of other abiotic and biotic factors, and their temporal consequences on the reflectance spectrum, could give insights to other similarities in the spectral and genetic structures of populations, and the genetically constrained processes that influence these similarities. For example, changes in leaf optical properties may be induced by increases in ambient carbon dioxide and ozone (Carter et al., 1995), aerosols (Farmer, 1993), or wind speed, light availability and thus irradiation that influence the water potential of plant tissue. Additionally, resistance to infections from bacteria, viruses and fungi, or attack by insects may also vary significantly on an intraspecific level and could be potentially derived from spectral signatures (eg. Carter, 1993). Correlative observations between genetic structure, reflectance spectra and environmental variables are indicators of physiological processes specific for a population adapted to its environment and under given environmental conditions.

While the above suggests that even more environmental conditions will likely affect spectral-genetic similarities, our proof-of-concept study was small and correlative, which at the same time limits confidence regarding the results for multiple spectral features and multiple environmental variables. Ideally, the spectra of a much larger number of replicated tree genotypes (or genetic groups such as seed families) should be analyzed across environmental variables that are not correlated with each other. This could be best done in experimental settings, which we now do with seed families collected from populations of *F. sylvatica* across Europe, or with very large numbers of trees in observational studies by multivariate matched sampling of pixels or trees (Imbens and Rubin, 2015, Sturm et al., 2022).

Further, we approach the analyses in a discrete manner by considering every time point and their environmental characteristic separately. Even though the multi-temporal selected acquisitions were distributed across growing seasons, the ontogenetic stages of the tree crowns were specific for particular years. Thus, segregation of time points by growing degree days does not allow us to follow the trajectories of individuals in time (day-to-day or year-to-year) (eg. Castro-Esau et al., 2006; Asner et al., 2006). Discovering such a trend could be facilitated by a long sequence of acquisitions on a more frequent basis that assess spectral information at regular time intervals (Cawse-Nicholson et al., 2021), although these are limited by instrument lifespan, cloud-free conditions and potentially coarser spatial resolution which would not directly support the assessment of individual trees and thus not help in detecting genetic variation within them. Further, such acquisitions are generally constrained to a particular time of the day as a practical consequence of satellite orbits, which would reduce the variation across acquisitions caused by systematic daily variations (eg. midday depression). From our analyses, we deduce that the mean reflectance spectra of the acquisitions over the day (six acquisitions during the 10th time point and five acquisitions during the 23th time point) may result in relatively low spectral-genetic similarities across the spectrum. Only distinct spectral ranges express relatively higher spectral-genetic similarities (500–650 nm and 700–1800 nm), which could be potentially stable across the day. Unraveling which of the spectral ranges are stable over the day and most informative about the genetic structure would require more within-day acquisitions. Here we did not analyze the within-day temporal variation because the missions within a day (time point) were not flown at consistently selected times of the day and because this additional possible analysis would have further increased the multiple-testing level for a small sample size of 68 trees.

When investigating multi-temporal datasets, it is especially important to consider differences in the particular geometry of the acquisition and the atmospheric composition. These could cause inherent variations in shading, directional effects (Schaepman-Strub et al., 2006) and atmospheric variation of the signal (Thompson et al., 2019). Combining multiple acquisitions from one day that vary in illumination/observation geometry to represent single time points potentially allows to average the variation due to illumination and observation geometries as well as atmospheric variability. By weighting the spectral information of the pixels by their inverse uncertainties, we account for the full set of spectral information and their relative reliability. Modifying reflectance spectra based on measurement uncertainties according to our results revealed up to 36% more information about biological variation. Still unknown is the impact of inclined surfaces on upright growing trees, their anisotropic behavior (Ma et al., 2020; Vögtli et al., 2020) and the uncertainties of surface reflectance spectra (Richter and Schläpfer, 2002; Carmon et al., 2020; Hueni et al., 2016).

It has to be noted that our proof-of-concept study is species-, site-, and time-specific. We show that prediction of the genetic structure is possible to a certain degree, with degree depending on airborne imaging spectroscopy measurements at specific time points, on combining shorter-term with longer-term observation schemes, and on specific ranges of wavelengths used. Our approach could be transferred to other species, different spatial and temporal scales, and different sensor systems as long as the observed biological units represent distinct genetic identities. In this study, we were able to investigate population-level genetic diversity for European beech trees in natural conditions by incorporating ground-based LiDAR (Morsdorf et al., 2018; Schneider et al., 2019), airborne laser scanning (Kaartinen et al., 2012; Schneider et al., 2014; Wang et al., 2016) and the 2 m-resolution pixel size of the airborne imaging acquisitions which allowed us to delineate *F. sylvatica* individuals (Guillén-Escribà et al., 2021) and to link them with their genetic identities. Working with data with a spatial resolution that is not suited to delineate individual trees or species would require community-based analyses, where the signal is averaged across several individuals and species (Zheng et al., 2021). As genetic differences between sites over larger areas are likely larger than genetic differences between individuals within a single site, this could still reveal genetic or phylogenetic variation at a cruder level. Finally, a better understanding of the mechanistic links between reflectance spectra, genetic and trait variation could be discovered by genome-wide association studies using e.g. whole-genome sequencing or gene capture datasets.

Our proof-of-concept study is only a first step to establish correlations between remotely sensed spectral information and genetic diversity in a dominant tree species. There is still a long way to go from this to predicting genetic diversity using spectral information in new areas where trees have not been genotyped. Prediction of genetic structure would require relationships calibrated for genetic resolution of interest and likely will need to account for spatially resolved environmental conditions. It would likely be supported by temporally rich acquisitions representing dynamic biological changes, and a well-understood impact of instrument variation on the spectral signal provided with its uncertainties as non-biological variation (Petibon et al., 2021). However, our study indicates wavelength bands which should be considered and under which environmental conditions multiple spectral signatures should be acquired to develop such prediction tools. Further studies will be needed to determine the degree to which calibrations obtained for one species in one area may be generalizable to other species and other areas.

5. Conclusions

This study demonstrates the potential of dense multi-temporal imaging spectroscopy data to link spectral information with genetic structure of temperate forest trees. We investigated 69 airborne imaging spectrometer data acquisitions along with genetic information from 68 tree individuals and environmental data to derive spectral-genetic similarities in population structure under different environmental conditions.

We observed that more genetically similar individuals have more similar reflectance in distinct spectral regions correlated with particular environmental conditions such as temperature and water availability on

different time scales. From this, we indicate traits and physiological processes that may explain these similarities and may correspond to genetic variation among tree individuals. We could demonstrate that relative differences between spectral bands were more informative about genetic structure than single-spectral band analyses, at least for specific times of the growing season. We were able to assess the quality of our derivation by incorporating uncertainty propagation at each step of analysis.

With dense multi-temporal observations of reflectance spectra indicative of multiple plant traits, we hope to contribute to the scalable monitoring of genetic diversity via remote sensing. That will allow us to better understand what is driving biodiversity changes, how to stop further biodiversity decline, and how to restore ecosystems where decline has taken place.

Author contribution

Ewa A. Czyż: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Visualization, Writing – original draft, Writing – review & editing. **Bernhard Schmid:** Conceptualization, Methodology, Writing – review & editing. **Andreas Hueni:** Data curation, Formal analysis, Writing – review & editing, Methodology. **Maarten B. Eppinga:** Conceptualization, Methodology, Writing – review & editing. **Meredith C. Schuman:** Writing – review & editing. **Fabian D. Schneider:** Methodology, Data curation, Writing – review & editing. **Carla Guillén-Escribà:** Data curation. **Michael E. Schaepman:** Conceptualization, Project administration, Resources, Funding acquisition, Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Datasets and scripts available from the Dryad Digital Repository: https://doi.org/10.5061/dryad.vdncjsxzk

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.rse.2022.113338.

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